Stefan Inzer

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Education

University of Washington
MS/PhD in Statistics *Sep 2024–Jun 2029*
- Preliminary exam: replicated Díaz and Hejazi (2020, JRSSB) on causal mediation analysis for stochastic interventions

University of California, Berkeley
BA in Applied Mathematics, Highest Distinction, Score: 4.0 *Aug 2020–Dec 2023*

Experience

Predoctoral Research Assistant Jun 2026–Present
NSF AMPS: Bayesian Physics-Informed Statistical Methods for Modern Power Systems
- Independently build physics-informed Bayesian models for power systems inference, advised by PI Abel Rodriguez
- Implement a Gibbs sampler in R (Galerkin prior, Hamiltonian Monte Carlo) to infer PDE solutions and parameters
- Run large-scale Monte Carlo simulations and build visualizations to test how well the posterior recovers the true parameters
- Analyze how regularization affects posterior spread around the true PDE parameters to guide model tuning

Teaching Assistant Sep 2024–Jun 2026
University of Washington
- Led weekly sections communicating statistical methods to students in STAT 311, 390, and 556
- Mentored two undergraduates through the SPA Directed Reading Program, guiding an independent study of time series from core concepts to ARIMA and GARCH models on economic data in R

Research Data Analyst (started as REU Student) May 2022–Apr 2025
UCLA AI and Cultural Heritage Lab
- Built an end-to-end Python NLP pipeline extracting semantic triples from 984 Holocaust survivor testimonies
- Adapted BERT and Bi-LSTM models for triple extraction, with results competitive against rule-based methods
- Flagged anomalous testimonies among 1,000 by modeling each transcript as a categorical time series of topics
- Used the OpenAI API to extract semantic triples and resolve ambiguous pronouns across interview transcripts

Statistics Research Intern Jun 2023–Aug 2023
IPAM: Research in Industrial Projects for Students (RIPS)
- Forecast GPS satellite orbit and clock error with a Gauss-Markov model in Python for The Aerospace Corporation
- Added L1 (lasso) regularization and benchmarked real-time orbit predictions against retrospective measurements
- Collaborated on the final report and presented our team's results to academic and industry researchers

Technical Skills

Programming: Expert **Keywords**: Python, R, C++, SQL, Git
Python Libraries: Advanced **Keywords**: pandas, NumPy, SciPy, scikit-learn, PyTorch
Statistics: Expert **Keywords**: Bayesian Inference, MCMC, Time Series, Causal Inference, Semiparametrics
Machine Learning and NLP: Advanced **Keywords**: Optimization, Deep Learning (BERT, LSTM), Topic Modeling, LLM APIs

Publications

Multi-scale Hybridized Topic Modeling Mar 2023
SIAM Undergraduate Research Online *https://www.siam.org/media/q30biptr/s153683r.pdf*
- Co-authored a multi-scale topic model (BERTopic + NMF) for survivor testimonies and 460,000+ news transcripts

Leadership and Activities

Treasurer and Filmmaker Aug 2020–Dec 2023
Cinematic Arts and Production Club (CAP), UC Berkeley
- Managed finances for Berkeley's largest student film organization, raising $4,500 and planning sponsorships and premieres
- Taught five editing workshops to 200+ members; collaborated on 12 short films as writer, director, and editor